Cluster-Based Identification of Role Archetypes from NBA Player Statistics

CSCI 4022

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Alex Schwab

Github: https://github.com/alsc5707/CSCI4022Project/tree/main
Youtube: https://youtu.be/Et1Edd9xaek?si=6wticgET6S8Ux4yw

1 Introduction

Modern NBA analytics are increasingly relying on quantitative performance metrics rather than typical positional averages. Although players are usually categorized as guards, forwards, or centers, the modern NBA emphasizes versatility and flexibility of role. This raises the question of whether players' statistical performances can naturally group them into new archetypes and whether these new archetypes align with their assigned position.

In this project, I analyzed NBA player performance statistics from the 2025-26 season using K-means clustering and principal component analysis (PCA) to identify groupings of players and evaluate what these groupings mean. Additionally, I used this cluster data to determine how many players are playing "out of position".

The data set used in this analysis was obtained from Kaggle and contains box score statistics for NBA players for every game. The main stats I focused on and used from the 25-26 season were:

- points
- assists
- offensive rebounds
- defensive rebounds
- steals
- blocks
- turnovers
- minutes played
- shooting percentages

Because this dataset is observational rather than experimental, the goal of this project is exploratory: to identify trends in player performance and evaluate whether statistical similarity aligns with position.

Specifically, this project addresses the following research questions:

- Can clustering identify distinct player archetypes?
- Do the derived clusters align with the listed player positions?
- How many players appear statistically "out of position"?

2 Methods and Results

2.1 Data Preparation

The dataset used in this project consists of NBA player performance statistics. The variables used in the analysis included scoring, passing, rebounding, defense, turnovers, playing time, and shooting efficiency.

First, since the data was per game data, I averaged each players stats across the 2025-26 season. I also cleaned the data. This was done with the following code:

```
    stats["gameDate"] = pd.to_datetime(stats["gameDate"])
df = stats[
    (stats["gameDate"] >= "2025-10-21") &
    (stats["gameDate"] <= "2026-04-12")
].copy()
#Regular season only
df = df[df["gameType"] == "Regular Season"]
numeric_cols = [
    "points",
    "reboundsOffensive",
    "reboundsDefensive",
    "assists",
    "steals",
    "blocks",
    "turnovers",
    "numMinutes",
    "fieldGoalsPercentage",
    "threePointersPercentage"
]
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors="coerce")
#remove DNP
df = df[df["numMinutes"] > 0].copy()
player_avg = df.groupby("personId").agg({
    "points": "mean",
    "reboundsOffensive": "mean",
    "reboundsDefensive": "mean",
    "assists": "mean",
    "steals": "mean",
    "blocks": "mean",
    "turnovers": "mean",
    "numMinutes": "mean",
    "fieldGoalsPercentage": "mean",
    "threePointersPercentage": "mean"
}).reset_index()
```

Because these variables are measured on different scales, the data was standardized before clustering. Standardization prevented variables with larger numerical ranges, such as points or minutes played, from dominating the clustering results. However, not every player was used. I wanted to conduct analysis on mostly relevant players, therefore I removed any player who had a minute average below 10 minutes. Then, I standardized the data.

Each value x_{ij} was transformed into a standardized value z_{ij} using

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where x_{ij} is the value of variable j for player i, μ_j is the mean of variable j, and σ_j is the standard deviation of variable j.

This was implemented in Python using `StandardScaler`:

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

2.2 Elbow Method

Before running K-means, I needed a K value. The elbow method was used to choose an appropriate number of clusters. This method compares the within-cluster sum of squares for different values of K.

The within-cluster sum of squares is calculated as

$$WCSS = \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

where K is the number of clusters, C_k is the set of players assigned to cluster k, x_i is the standardized statistic vector for player i, and μ_k is the centroid of cluster k.

This was implemented by running K-means for several possible values of K and storing the inertia value:

```
inertias = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertias.append(kmeans.inertia_)
```

The elbow plot was then used to identify a value of K where adding more clusters gave diminishing improvements.

As we can see in the following graph, $k = 4$ is where the line bends, so I used a k of 4.

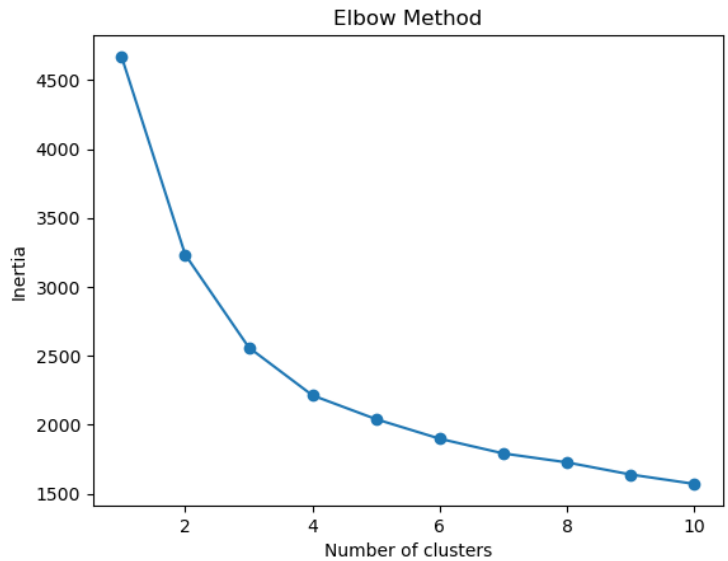

Figure 1: Elbow plot used to compare K-means inertia values for different numbers of clusters.

2.3 K-Means Clustering

After selecting the number of clusters, K-means clustering was applied to the standardized data. K-means assigns each player to the nearest cluster centroid and updates the centroids until the cluster assignments stabilize.

The K-means objective is to minimize

$$\sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2$$

where x_i is the standardized feature vector for player i and μ_k is the centroid of cluster k.

The distance between a player and a cluster centroid was measured using Euclidean distance:

$$d(x_i, \mu_k) = \sqrt{\sum_{j=1}^{p} (x_{ij} - \mu_{kj})^2}$$

where p is the number of variables used in the clustering model.

This was implemented using the `KMeans` function from `scikit-learn`:

```
kmeans = KMeans(n_clusters=4, random_state=42)
data["cluster"] = kmeans.fit_predict(X_scaled)
```

The resulting clusters represented groups of players with similar statistical profiles. These groups can be interpreted as player archetypes, such as scoring-oriented players, defensive players, playmakers, and interior players.

2.4 Principal Component Analysis for Visualization

After clustering, Principal Component Analysis was used to reduce the standardized data to two dimensions so the clusters could be visualized.

PCA finds directions of maximum variation in the standardized data. The covariance matrix of the standardized data matrix Z is

$$S = \frac{1}{n-1} Z^T Z$$

where n is the number of players.

The principal component directions are found by solving

$$Sv_k = \lambda_k v_k$$

where v_k is the kth principal component direction and λ_k is the variance explained by that component.

Each player was projected onto the first two principal components using

$$Y = ZV$$

where Y is the matrix of PCA scores and V contains the principal component vectors.

The amount of variance explained by each principal component was calculated as

$$ExplainedVarianceRatio_k = \frac{\lambda_k}{\sum_{j=1}^{p} \lambda_j}$$

PCA was implemented in Python as follows:

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

data["PC1"] = X_pca[:, 0]
data["PC2"] = X_pca[:, 1]
```

The PCA scores were then used to create a scatter plot of the clusters:

```
plt.figure(figsize=(8,6))
plt.scatter(
    data["PC1"],
    data["PC2"],
    c=data["cluster"]
)
```

```
plt.title("NBA Player Clusters")
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.show()
```

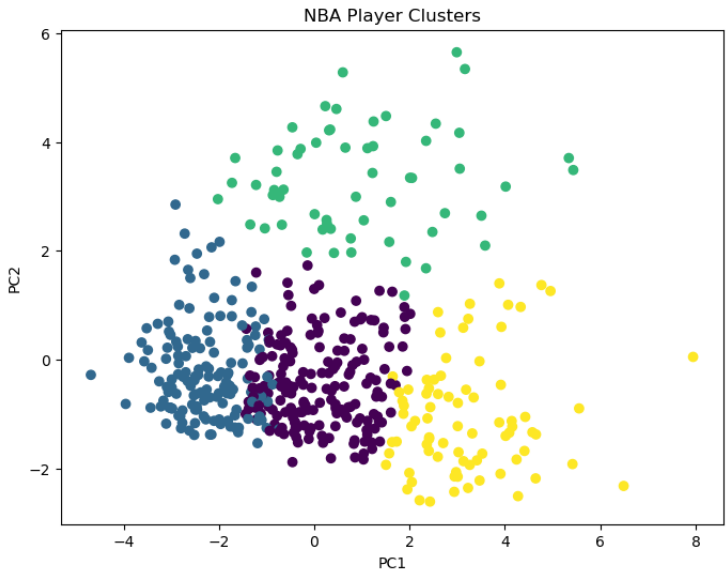

Figure 2: K-means cluster assignments visualized using the first two principal components. PCA was used only for visualization after clustering.

2.5 Position Alignment Analysis

To evaluate whether the clusters corresponded to traditional basketball positions, each player's cluster assignment was compared with their listed position.

For each cluster, the most common listed position was identified and treated as the expected position for that cluster. Players whose listed position differed from the expected position of their cluster were identified as out-of-position players.

This was implemented using a cross tabulation:

```
cluster_position_table = pd.crosstab(
    data["cluster"],
    data["position"]
)

expected_position = cluster_position_table.idxmax(axis=1)
```

```
data["expected_position"] = data["cluster"].map(expected_position)

out_of_position_players = data[
    data["position"] != data["expected_position"]
]
```

This comparison helped determine whether traditional position labels matched the statistical groups found by K-means.

2.6 Interpretation of Results

The clustering results showed that NBA players can be grouped into meaningful statistical archetypes based on performance data. Some clusters aligned closely with traditional positions, while others contained players with more flexible roles.

The PCA visualization helped show these clusters in two dimensions. Players located close together on the PCA plot had similar overall statistical profiles, while players farther apart differed more strongly in their production patterns.

As a result of the clustering, the following tables were created. The first demonstrates the averages of each cluster, and the second the position labels for each player in the clusters.

Cluster	Points	Off Reb	Def Reb	Assists	Steals	Blocks	TO	Minutes	FG%	3PT%
0	11.03	0.90	2.93	2.39	0.88	0.42	1.23	24.59	0.44	0.33
1	5.49	0.68	1.70	1.40	0.51	0.23	0.75	14.81	0.37	0.24
2	9.96	2.68	4.70	1.72	0.73	1.04	1.27	22.74	0.58	0.09
3	20.35	1.11	4.55	5.36	1.19	0.49	2.54	31.53	0.47	0.33

Table 1: Average performance statistics for players in each cluster.

Cluster	Center	Forward	Guard
0	6	70	107
1	9	54	83
2	32	26	1
3	3	25	51

Table 2: Number of players from each listed position within each cluster.

By analyzing the results from each cluster, 4 archetypes of players were created to associate the players to. These clusters represent different roles within modern NBA team structure rather than strictly traditional position labels.

- **Cluster 0: 3-and-D Wing Players**
  - Players in this cluster contribute moderate scoring production while maintaining strong perimeter shooting efficiency, particularly from three-point range.
  - These players generate defensive impact through steals rather than shot blocking, suggesting perimeter defensive responsibilities.
  - They play consistent rotational minutes, indicating regular inclusion in starting or primary bench lineups.
  - The position distribution shows this cluster is dominated by guards and forwards, supporting the interpretation of versatile wing defenders.
  - Overall, this cluster represents role players who space the floor offensively while contributing perimeter defense.
- **Cluster 1: Low-Usage Rotational Guards**
  - Players in this cluster record the lowest averages in scoring, assists, and minutes played across all clusters.
  - These players handle the ball less frequently and contribute fewer counting statistics overall.
  - The lower playing time suggests they primarily serve second-unit or situational bench roles.
  - Their statistical profile reflects support responsibilities rather than primary offensive creation.
  - This cluster is largely composed of guards and forwards with limited offensive usage rates.
- **Cluster 2: Interior Anchors**
  - Players in this cluster produce the highest rebounding and shot-blocking averages among all clusters.
  - Their strong field goal percentage reflects scoring efficiency near the basket rather than perimeter shooting.
  - These players attempt very few three-point shots, reinforcing their interior offensive role.
  - Assist totals remain relatively low, indicating limited playmaking responsibilities.
  - The position distribution confirms that this cluster is dominated by centers, supporting the interpretation of rim protectors and interior defenders.

- **Cluster 3: Primary Shot Creators**
  - Players in this cluster lead all groups in scoring, assists, and minutes played.
  - Their elevated assist production indicates that they function as offensive initiators rather than purely scoring specialists.
  - These players control possessions more frequently and serve as primary decision-makers within offensive systems.
  - The combination of scoring and playmaking identifies this cluster as the group most responsible for offensive creation.
  - The position distribution includes both guards and forwards, reflecting modern positionless offensive leadership roles.

After analyzing clusters, a comparison on the positions of the players was done. It was found that roughly 194 out of 580 players were classified as "out of position". This is equal to roughly $\frac{1}{3}$ (.3345) of all unique players who played this season. Additionally, after analyzing the players listed as "out of position" it was found that a majority of these players were role players and/or bench players. Logically, this makes sense. If a team is requiring a player play a different role than they are used to, it is more likely that they ask a non superstar, non first-option player to adapt their play style. This also makes sense for bench players who are fighting for a starting role, and feel like they may need to branch out to other areas of expertise to earn that spot.

3 Conclusion

This project applied K-means clustering and Principal Component Analysis (PCA) to NBA player performance statistics in order to identify player archetypes and evaluate how well these archetypes align with traditional position labels. The clustering results revealed four distinct player role groups that reflect modern basketball responsibilities more accurately than positional labels alone.

Cluster 0 represents **3-and-D wing players**, who contribute moderate scoring production while maintaining strong three-point shooting efficiency and generating defensive impact through perimeter activity such as steals.

Cluster 1 corresponds to **low-usage rotational guards**. Players in this group record the lowest averages in scoring, assists, and minutes played.

Cluster 2 represents **interior anchors**. These players produce the highest rebounding and shot-blocking averages and rarely attempt three-point shots.

Cluster 3 corresponds to **primary shot creators**. These players lead all clusters in scoring, assists, and minutes played, indicating that they function as offensive initiators.

A comparison between statistical clusters and listed player positions showed that traditional positional labels often align with statistical archetypes but do not fully capture player roles. Several players were assigned to clusters that

differed from their listed position, suggesting that performance-based role identification provides additional insight beyond position classification alone. Out of 580 unique players that played in the 2025-26 season, 194 were categorized as "out of position", equating to roughly $\frac{1}{3}$ of all players.

Principal Component Analysis was used to visualize the clustering structure in two dimensions. The PCA projection demonstrated clear separation between out clusters, confirming that the clustering algorithm identified meaningful structure within the dataset.

4 Limitations and Future Work

There are several limitations to this project. First, the dataset is observational, so the results show patterns and relationships but do not prove causation. Second, the analysis only uses box score statistics, which may not fully capture defensive impact, team strategy, or off ball movement. A player could be the best at defensive pressure but have no box score stats to back that up, and thus in this analysis that nuance is lost. These stats would be more accurately tracked in advanced statistics that were not used.

Another limitation is that clustering assigns each player to one unique group, even though many players have hybrid roles. In reality, player roles exist on a spectrum rather than in perfectly separate categories.

Future work could improve this analysis by including multiple seasons of data, advanced statistics, and team context. Additional methods could also be applied, such as Gaussian mixture models, to see if a different result is drawn.

A future version of this project could also compare results across seasons to see whether a player's role change over time.

5 References

- PlayerStatistics.csv, NBA Dataset: Box Scores and Stats (1947 - Today), Kaggle, https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores?select=PlayerStatistics.csv
- Players.csv, NBA Dataset: Box Scores and Stats (1947 - Today), Kaggle, https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores?select=Players.csv
- CSCI 4022 Project, https://github.com/alsc5707/CSCI4022Project
- Project Presentation Video, https://youtu.be/Et1Edd9xaek?si=6wticgET6S8Ux4yw